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MedSpace AI — New Medical AI × Knowledge Graph Website & 5-Product Plan

Working domain candidates: ``medspace.ai`` (fits the **space** family),
``spacemed.ai``, ``medgraph.ai``, ``clinicspace.ai``
Stack: Next.js (same GitHub → deploy pipeline as the other 8 sites) ·
Backend: VeritasGraph + ``clinical-kg`` FastAPI · Local models via Ollama (HIPAA
on-prem) or cloud API tier.

> **Positioning (one line):** *"The knowledge-graph AI layer for clinics —
connects your patients, doctors, devices and records into one governed graph,
with a verifiable source citation on every answer."*

Why VeritasGraph is the moat: every product below returns answers with
span-level `[doc#chunk]` citations, a reasoning path, and runs 100% on-
prem (``clinical-kg`` pipeline: de-identify → extract → reconcile → normalize
to ICD-10/SNOMED/RxNorm/LOINC → graph → cited answer). In healthcare,
auditable AI is the product — not the model.

1. Market: what clinics are actually paying for (highest demand, fastest
ROI)

Rank	Pain / demand	Who pays	Why it's urgent
1	Clinical documentation burden (doctors spend ~2 hrs on notes per 1 hr of care)	Doctors, clinic owners	Burnout, fewer patients/day → direct revenue loss
2	Remote patient monitoring (RPM) + chronic care	Clinics (billable: CPT 99453/99454/99457/99458 in US; equivalent chronic-care programs elsewhere)	Recurring reimbursed revenue, hardware-linked
3	No-shows, front-desk load, patient follow-up	Clinic management	10–30% no-show rate = idle chairs/rooms
4	Coding / billing errors & denials	Clinic management	5–10% claims denied; recoverable money
5	Patient wants answers 24/7 (symptoms, meds, reports)	Patients → clinic brand	Retention + trust; drives new client acquisition

The 5 products below map 1:1 to this table, share **one knowledge graph**, and are sold as a **ladder** — a clinic can start with one at \$/month and add the rest without new integration.

2. The 5 products

Product 1 — **MedScribe Graph** • **AI Clinical Documentation (for Doctors)**

- **What:** Ambient / dictated consultation → structured SOAP note → auto-coded (ICD-10, SNOMED, RxNorm) → written into the patient's knowledge graph with citations back to the transcript span.
- **Hardware link:** any phone/tablet mic, room microphone array, or dictation pen; runs on a local GPU box (or Ollama on clinic PC) for zero data egress.
- **Built from:** `clinical-kg` extract + normalize + assertion reconciler (contradiction detection is a unique selling point: *"HPI says no diabetes, problem list says T2DM — flagged"*).
- **ROI to clinic:** saves ~1.5–2 hrs/doctor/day → 2–4 extra visits/day. Pays for itself with **one extra consultation per week**.
- **Pricing:** per-doctor/month + one-time on-prem box.

Product 2 — **VitalGraph RPM** • **Remote Patient Monitoring & Chronic Care (Hardware-linked)**

- **What:** Connects BP cuffs, glucometers, pulse-ox, smart scales, ECG patches, wearables (Bluetooth / BLE gateway / FHIR Observation) → vitals stream fused with the patient's clinical graph → risk scores + explainable alerts to the doctor, plain-language nudges to the patient.
- **Hardware link:** Off-the-shelf BLE devices + a small clinic/home gateway (Raspberry Pi / Android box) or direct phone pairing. Vendor-agnostic device catalog.
- **Built from:** **MedKG-Signal** (`medkg-signal-reference`) — vitals + text + graph fusion; validated AUROC 0.83 in-hospital mortality (PhysioNet 2012), 0.93 on EHR cohort. Explanation contract = KG evidence paths, i.e. every alert is citable.
- **ROI to clinic:** **new recurring reimbursed revenue** (RPM/CCM billing) + fewer readmissions/ER visits. This is the highest-ROI product in the ladder and the strongest hardware story.
- **Pricing:** per-monitored-patient/month + hardware kit margin.

Product 3 — **ClinicOps Graph** • **Front-Desk, Scheduling & Revenue AI (for Clinic Management)**

- **What:** AI receptionist (WhatsApp/Telegram/web/phone) for booking, reminders, rescheduling, intake forms; no-show prediction; automated claim-coding check from the graph (Product 1 output) to cut denials; management dashboard (utilisation, revenue/room, doctor throughput).
- **Hardware link:** self-check-in kiosk/tablet at reception, queue display screens (reuse **space-sign.ai** signage), room occupancy sensors (reuse **spacelizit** / **spacelean** space analytics).
- **Built from:** `Telegram-MedicalBot` + VeritasGraph Studio agents (guardrails, tools, memory) + `spacecrm.net` CRM primitives.
- **ROI to clinic:** -30–50% no-shows, -1 FTE front desk, +recovered denied claims. Simple to demonstrate in a 2-week pilot.
- **Pricing:** per-clinic/month, tiered by rooms/doctors.

Product 4 – **PatientGraph Companion** • Patient-Facing Health Assistant (for Patients)

- **What:** Patient app/portal: upload lab reports & prescriptions → plain-language explanation **with citation to the report line**; medication reminders & interaction checks (RxNorm graph); symptom triage that routes to *this clinic's* booking (feeds Product 3); post-visit follow-up.
- **Hardware link:** pulls from the same RPM devices (Product 2) so the patient sees their own trends.
- **Built from:** VeritasGraph GraphRAG + guardrails (medical-safety policy rules in `rules/`), `openmed` de-ID so PHI never leaves the clinic.
- **ROI to clinic:** patient retention, referral growth, fewer phone calls; **it is the acquisition funnel** — a free/basic tier brings patients, patients bring clinics.
- **Pricing:** free tier for patients (clinic-branded) → premium; clinic pays white-label fee.

Product 5 – ****ClinicalKG Enterprise**** • Governed Knowledge Graph & Research/Compliance Platform (for Hospitals, Chains, Insurers, Researchers)

- **What:** The full `clinical-kg`` on-prem platform: de-identification vault, contradiction detection, cohort multi-hop queries (*"patients on metformin with eGFR < 45 and no nephrology referral"*), k-anonymity / re-ID risk scoring, audit trail, FHIR/HL7 import, MCP server for other tools.
- **Hardware link:** ships as an on-prem appliance / GPU server or private cloud; air-gapped mode.
- **Built from:** `clinical-kg`` + `veritasgraph_mcp`` + `veritas-reason`` policy engine + Studio.
- **ROI:** research cohort discovery in minutes vs weeks; compliance audit readiness; quality-measure reporting; feeds all 4 products above for multi-site groups.
- **Pricing:** annual enterprise license + implementation.

How they connect (one graph, five doors)

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    Patients → P4 Companion ┌
    Devices → P2 VitalGraph ┤
    Doctors → P1 MedScribe ┤ → VeritasGraph Clinical Knowledge Graph (on-
    prem, cited) → P5 Enterprise
    Front desk ▶ P3 ClinicOps └
                                ▲ de-ID • ICD-10/SNOMED/RxNorm/LOINC •
    contradictions • citations

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Each product is useful alone; each one adds data that makes the others smarter. That's the upsell engine.

3. "Easy to incorporate" – integration design

- ****No rip-and-replace:**** works **beside** any EHR/PMS. Import via FHIR R4, HL7 v2, CSV/PDF upload, or just the microphone. Export back as FHIR/PDF.
- ****3 deployment tiers:**** ① cloud (fast pilot, de-identified) ② hybrid (PHI stays local, LLM optional cloud) ③ fully on-prem/air-gapped (hospitals). Same codebase — same as VeritasGraph ``lite / local / full`` modes.
- ****Hardware kits:**** "RPM Starter Kit" (gateway + BP + glucometer + scale), "Front-Desk Kit" (kiosk tablet + display), "Clinic AI Box" (mini GPU PC with Ollama preloaded). Plug-and-play, pre-paired.
- ****Channels patients already use:**** WhatsApp, Telegram, SMS, web — no app download needed to start.
- ****MCP + REST API**** so partners/system integrators can extend.

4. "Clients from any level" — go-to-market ladder

Level	Entry product	Entry price feel	Path
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Solo doctor / small clinic	P1 MedScribe or P3 ClinicOps	low monthly, card payment, self-serve	→ add P2 when they see RPM revenue
Multi-doctor clinic / polyclinic	P3 + P2	per-clinic bundle + hardware kit	→ P1 for all doctors → P4 branded app
Hospital / chain / insurer	P5 Enterprise	annual license	→ roll out P1–P4 across sites
Patients (B2C funnel)	P4 free tier	free	→ they ask their clinic → clinic buys
Device makers / SI partners	White-label P2 / API	revenue share	→ distribution

Acquisition tactics: free ROI calculator on the site, 14-day pilot with a real clinic metric (no-show rate, notes time), country-specific reimbursement guides (US RPM/CCM, UAE/India/EU chronic-care programs), referral commission for doctors.

5. Website structure (Next.js)

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/	Hero: "One knowledge graph. Five clinic superpowers. Every answer cited."		
/products/medscribe	P1		
/products/vitalgraph	P2 (+ device compatibility list)		
/products/clinicops	P3		
/products/companion	P4 (patient landing, app links)		
/products/enterprise	P5		
/hardware	Kits, gateways, supported devices, "Clinic AI Box"		
/roi-calculator	Interactive: doctors × visits × no-show% → \$ saved/yr		
/for-doctors	/for-clinics	/for-patients	/for-hospitals persona pages
/integrations	FHIR, HL7, EHR list, WhatsApp/Telegram, MCP		

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/security          HIPAA / GDPR / on-prem, de-ID vault, audit trail
/technology        "Powered by VeritasGraph" – knowledge graph + source
attribution
/evidence          Benchmarks (AUROC 0.93 / 0.83), papers, citations
/pricing           Ladder tiers
/blog  /docs  /llms.txt
^^^
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6. GEO/SEO – plug into the existing playbook

Apply everything in [veritasgraph-geo-seo-website-integration.md]
(veritasgraph-geo-seo-website-integration.md) plus medical-specific items:

- **Schema:** ``MedicalOrganization`/`SoftwareApplication``
(``applicationCategory: "MedicalApplication"``), ``MedicalWebPage``, ``FAQPage``,
``Product`` + ``Offer`` for hardware kits, ``isBasedOn`` → VeritasGraph.
- **E-E-A-T:** author every clinical page with a named clinician reviewer +
``reviewedBy``; cite the ICSPIS/MedKG-Signal paper and PhysioNet benchmarks on
``/evidence`` – this is exactly the citable content generative engines quote.
- **Keyword clusters:** **AI clinical documentation software*, *remote patient
monitoring platform*, *AI receptionist for clinics*, *clinical knowledge
graph*, *HIPAA on-prem AI*, *explainable medical AI*, *lab report explainer*,
no-show prediction.*
- **GEO hooks:** FAQ answers to **"Which AI scribe keeps data on-premise?"*,
"How does RPM billing work?", *"Is the AI's medical answer verifiable?"** –
each answered with a VeritasGraph citation sentence.
- **Cluster linking:** the new site joins the entity cluster – links to
VeritasGraph, ``bibinprathap.com``, and reuses ``space-sign.ai`` (signage),
``spacecrm.net`` (CRM), ``spacelizit/spacelean`` (room analytics) as integrations
→ reciprocal backlinks.
- ``/llms.txt`` listing the 5 products, VeritasGraph as the framework, and the
evidence page.

7. Compliance & safety guardrails (non-negotiable for trust and for selling)

- Patient-facing AI is **decision support, not diagnosis** – disclaimers +
red-flag escalation to a human/emergency.
- On-prem de-identification (``openmed`/`clinical-kg`` Safe Harbor + sealed
vault); PHI never in prompts to external APIs unless hybrid tier with BAA.
- Audit log for every AI output (who asked, which sources cited, model
version).
- Regulatory posture: P1/P3/P4 as non-device clinical workflow/admin software;
P2 alerts positioned as monitoring/triage with clinician-in-loop; document
intended use per market (FDA/CE/MDR/CDSCO). Consult regulatory counsel before
claims.

- Roles/consent: patient consent flows, role-based access (``veritas-reason`` policy rules).

8. Build order (MVP → revenue fastest)

1. ****Site + ROI calculator + waitlist**** (1 sprint) – start SEO/GEO indexing immediately.
2. ****P3 ClinicOps**** (Telegram/WhatsApp bot + booking + reminders) – quickest pilot win, no clinical risk, existing bot code.
3. ****P1 MedScribe**** – reuse ``clinical-kg`` pipeline; biggest doctor delight.
4. ****P2 VitalGraph RPM**** – hardware kit + MedKG-Signal; unlocks recurring reimbursed revenue.
5. ****P4 Companion**** – patient funnel on top of P2/P3 data.
6. ****P5 Enterprise**** – package what already runs for hospitals.

9. Reusable assets already in this repo

Product	Existing code
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P1, P5	<code>`clinical-kg/`</code> (backend pipeline + Next.js frontend, 34 tests)
P2	<code>`medkg-signal-reference/`</code> (fusion models, PhysioNet loaders, eval)
P3	<code>`Telegram-MedicalBot/`</code> , <code>`studio_api/`</code> (agents, tools, guardrails)
P4	<code>`veritasgraph/`</code> GraphRAG, <code>`rules/`</code> policy, <code>`openmed/`</code> de-ID
All	<code>`veritasgraph_mcp/`</code> , <code>`veritas-reason/`</code> , <code>`Awesome-AI4Med/`</code> , <code>`GNN_for_EHR/`</code> , ICSPIS paper in <code>`docs/`</code>